SafeMeter

A Vehicle-Powered Smart Speedometer That Detects Crashes and Summons Help in the Golden Hour

Team Name	[Your team name]	Entry Level	Open - Undergraduate
Theme	Surakshit Bharat	Sub-theme	[Select the road-safety / emergency-response sub-theme]
Track	Track A: Product or Technology	Prototype	Working hardware unit, volunteer app, dashboard

1. Executive Summary

SafeMeter is a low-cost smart speedometer for two-wheelers, autos, cabs and trucks. It replaces or sits beside the existing speedometer, shows live speed, and continuously monitors for crashes. When a crash is confirmed, it gives the rider 15 seconds to cancel, and if there is no cancellation it automatically sends an SMS with live location, speed at impact and a map link to family, registered nearby volunteers and emergency services. No internet connection, smartphone or user action is needed.

The core innovation is **sensor fusion**: instead of relying on an accelerometer alone (which confuses potholes with crashes), SafeMeter combines wheel speed, impact force and vehicle orientation. This cuts false alarms sharply while still detecting real crashes. A community of trained volunteers forms the response layer, so help does not depend only on ambulance availability.

2. Problem Statement

Road accidents are a major cause of death and serious injury in India, and a large share of victims are two-wheeler riders. Medical literature widely recognises a *golden hour* after trauma in which timely care greatly improves survival. Delays happen because:

- Nobody may be nearby to call for help, especially on highways, at night or in rural areas.
- Bystanders often hesitate to get involved, and the victim may be unconscious.
- The exact location is hard to communicate to emergency services.
- Existing solutions depend on a charged phone, an installed app or a helmet worn and charged correctly, so they fail in exactly the situations that matter.

[Add 1-2 verified statistics here, e.g. annual road deaths and the two-wheeler share, from the latest MoRTH report, with the source and year.]

3. Proposed Solution

SafeMeter is a vehicle-mounted unit that does four things: **measure** speed and motion, **decide** whether a crash has happened, **alert** people who can help, and **record** the incident so authorities can fix dangerous spots.

Stage	What happens	Technology
1. Sense	Wheel speed, impact force, tilt and GPS position are read continuously.	Hall sensor, MPU6050 IMU, NEO-6M GPS
2. Decide	A three-stage state machine confirms a crash and rejects potholes and hard braking.	ESP32 firmware, sensor fusion
3. Cancel window	Buzzer and display show a 15-second countdown so a minor fall or false alarm can be cancelled.	OLED display, buzzer, button

Stage	What happens	Technology
4. Alert	SMS with location link, speed at impact and time goes to contacts, volunteers and emergency services.	SIM800L GSM module (works without internet)
5. Respond	Nearby volunteers get an app notification; the first to accept becomes the responder.	Volunteer app with Firebase
6. Analyse	Incidents are logged on a dashboard with response times and black-spot heatmaps.	Web dashboard

4. System Architecture

On-vehicle unit: ESP32 microcontroller (main processor) with an MPU6050 accelerometer and gyroscope, a hall sensor and magnet on the wheel for true speed, a NEO-6M GPS module for location and backup speed, a SIM800L GSM module for SMS, an OLED display for speed and alerts, a buzzer and cancel button, and a buck converter powering everything from the vehicle battery.

Cloud and apps: The unit sends the alert to emergency contacts by SMS and, where connectivity exists, to a Firebase backend. The backend finds registered volunteers within a set radius and pushes a notification to the volunteer app. The first volunteer to accept is assigned, and others are told the case is taken. All events are stored for the dashboard.

Why SMS first: A large part of the road network has patchy mobile data but still has basic voice and SMS coverage. SMS keeps the life-critical alert working where internet-based apps would fail.

5. Crash Detection Design (Accelerometer and Sensor Fusion)

5.1 Sensors and settings

- **MPU6050 (prototype):** low-cost accelerometer and gyroscope. Configured at +/-16 g range, 100-200 Hz sampling over I2C, with the built-in low-pass filter (about 40 Hz) to remove engine vibration.
- **Production upgrade:** a high-g sensor such as ADXL375 or H3LIS331DL (about +/-200 g) to capture real crash magnitudes, which can exceed the prototype sensor range.
- **Mounting and calibration:** rigid mounting to the frame or speedometer housing. At power-up, the resting gravity vector is stored as the upright reference. Impact detection uses the total magnitude, $\text{sqrt}(ax^2 + ay^2 + az^2)$, so it does not depend on mounting orientation.

5.2 Distinguishing events

Event	Acceleration pattern	Wheel speed	After the event
Pothole	Short vertical spike, under about 100 ms	Continues	Normal riding
Speed breaker	Smaller, repeated spikes	Dips slightly, recovers	Normal riding
Hard braking	Sustained 0.5-0.8 g along travel direction, no spike	Falls gradually	Normal riding
Crash	**Large spike with high vibration**	**Collapses within about 1 s**	**Tilted or stationary**

5.3 Three-stage detection logic

- **Stage 1 - Trigger:** total acceleration crosses an impact threshold (starting value about 4 g for bench testing, tuned from collected data).

- **Stage 2 - Speed confirmation:** speed was above about 20 km/h before impact and fell by more than about 60% within 1 second. A pothole fails this test because speed continues.
- **Stage 3 - Orientation confirmation:** for two-wheelers, tilt beyond about 60 degrees for more than 2 seconds (vehicle on its side). For four-wheelers, no motion for several seconds.

Only when all three stages pass does the 15-second cancel countdown begin. Tilt is estimated with a complementary filter (about 98% gyro, 2% accelerometer), which is lightweight and runs easily on the ESP32.

5.4 Optional machine-learning layer

Labelled test data (potholes, braking, simulated crashes) will be used to train a small decision tree or lightweight classifier, compared against hand-set thresholds. Results will be reported honestly, including cases where thresholds perform equally well.

6. Volunteer Response Network

SafeMeter treats response as seriously as detection. Local volunteers (students, shopkeepers, riders' clubs, trained ward or panchayat members) register in the app with their location and availability. On an alert, the nearest available volunteers are notified, the first to accept sees the live map link and speed at impact, and the system records time-to-accept and time-to-arrival.

- **Training:** a short module covering safe scene approach, calling 108, and not moving suspected spinal-injury victims.
- **Safety and trust:** volunteers are registered with basic verification, and the app provides guidance rather than medical instructions.
- **Fallback:** alerts always go to family and emergency numbers in parallel, so the system never depends on volunteers alone.

7. Prototype and Demonstration Plan

7.1 Hardware bill of materials (prototype)

Component	Purpose
ESP32 development board	Main controller with Wi-Fi/Bluetooth
MPU6050 IMU	Impact force and tilt
Hall sensor and magnet	Accurate wheel speed
NEO-6M GPS	Location and backup speed
SIM800L GSM module	SMS alerts without internet
OLED/TFT display, buzzer, button	Speed display, countdown, cancel
Buck converter, enclosure	Vehicle power and housing

Estimated prototype cost is about Rs. 2,000-2,500 (to be confirmed with current prices). At 1,000 units, bulk sourcing and a custom board are expected to bring this towards Rs. 1,500. **[Replace with your actual costed figure.]**

7.2 Live demo flow (about 2 minutes)

- Mount the unit on a bike or a bench rig with a spinning wheel; the display shows live speed.
- Simulate a pothole: no alarm is raised.
- Simulate a crash: the countdown starts. Show cancelling once, then let it run out.

- The SMS arrives on a phone and the volunteer app pings with a map.
- The dashboard logs the incident with time, location and speed.

Safety note: all crash simulations use bench rigs or padded, controlled drop tests. No real vehicle crash or rider risk is involved.

8. Testing and Validation Plan

Raw sensor data will be logged and labelled for each trial, then summarised in an accuracy table.

Test category	Minimum trials	Metric reported
Pothole / speed breaker passes	20+	False alarm rate
Hard braking events	15+	False alarm rate
Simulated crashes (bench / padded drop)	15+	Detection rate, detection time
Alert delivery (SMS and app)	10+	End-to-end delay
Volunteer pilot (10-15 volunteers, test alerts)	5+ drills	Time to accept and reach

[After testing, insert your actual results here, e.g. detection rate, false alarm rate and average alert delay.]

9. Impact and Scalability

- **Faster help:** automatic alerts remove the dependence on bystanders and speak the victim's exact location within seconds.
- **Inclusive:** works on any vehicle, in any language, with no smartphone or data plan.
- **Data for prevention:** aggregated, anonymised incident data shows accident black spots so municipalities and traffic police can act.
- **Scalable:** starts as a retrofit unit for cabs, delivery fleets and two-wheelers, with a path to OEM integration. Fleet operators and insurers are natural early customers because faster response and driver safety reduce their costs.
- **Policy fit:** aligns with national road-safety priorities and emergency-response initiatives.

Possible business model

- Unit sale or fleet subscription (device plus alert service).
- Fleet and insurer partnerships for bulk deployment.
- Anonymised black-spot analytics offered to government bodies.

10. Innovation and Differentiation

Existing approach	Limitation	SafeMeter advantage
Phone crash-detection apps	Need charged phone, app running, data	Vehicle-powered, always on, SMS-based
Smart helmets	Depend on charging and wearing	No rider action needed
Accelerometer-only detectors	Confuse potholes with crashes	Speed + impact + tilt fusion
Emergency calls (108)	Need someone to call	Automatic alert plus volunteer layer

11. Limitations and Risks (and How We Address Them)

- **Cannot tell if the rider was thrown or injured:** addressed by the cancel window, and by sending speed at impact so responders can judge severity.
- **Bench tests produce lower forces than real crashes:** thresholds are prototype values; production would use a high-g sensor with field calibration.
- **Coverage gaps:** in zero-signal areas the unit stores the alert and sends when signal returns.
- **Installation and recurring SMS cost:** positioned as a retrofit unit; alert-only SMS plans are low cost per vehicle.
- **Privacy:** location is shared only when an alert fires; dashboard data is anonymised.

12. Implementation Timeline

Week	Activities
1	Assemble hardware, wire sensors, log raw data
2	Build detection logic, run 50+ labelled tests
3	Build volunteer app, SMS flow and dashboard
4	Enclosure, cost sheet, volunteer pilot, demo video

13. Conclusion

SafeMeter closes the most dangerous gap in road safety: the minutes between a crash and the arrival of help. By fusing wheel speed, impact and orientation, it detects crashes reliably without a phone or charging habit, and by pairing detection with a volunteer network and an incident dashboard, it turns a single alert into a faster, more accountable emergency response system.

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